

Derivatives

Kisi bhi point par function kitni tezi se badal raha hai

01 Opening Hook

STORY TIME

Socho tum ek pahaad chadh rahe ho. Kabhi steep chadhai hai, kabhi flat rasta. Har point par 'kitna steep hai' yeh batana – yehi hai **Derivative**.

02 Why Does This Concept Exist?

THE PROBLEM

AI training ka poora goal hai 'loss ko kam karna'. Iske liye humein pata hona chahiye ki har weight ko badalne se loss kis direction mein aur kitni tezi se change hoga. Derivative yeh exact jaankari deta hai – yeh 'sabse steep neeche jaane wala rasta' dhoondhne mein madad karta hai.

03 Explain Like I'm 10

SIMPLE WORDS

Socho tum ek car mein ho aur speedometer dekh rahe ho. Speed batati hai position kitni tezi se badal rahi hai time ke saath – yeh position ka derivative hai. Waise hi, loss function ka derivative batata hai loss kitni tezi se badal raha hai weight ke saath.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

Derivative ek function ke graph ka 'slope' hai kisi bhi point par. Positive slope matlab function badh raha hai, negative slope matlab ghat raha hai, zero slope matlab woh point flat hai (minimum, maximum, ya saddle point ho sakta hai).

05 Ab Thoda Maths

FORMULA TIME

Notation:

$f'(x)$ ya df/dx - dono derivative represent karte hain.

Power Rule:

$d/dx(x^n) = n \cdot x^{(n-1)}$. Jaise $d/dx(x^2) = 2x$.

06 Visualize Everything

PICTURE IT

$$f(x) = x^2$$

Slope (derivative) alag points par:

$\nearrow$ <- steep positive slope (bada derivative)
 $\searrow$ <- steep negative slope

07 AI Connection

WHERE AI USES THIS

Gradient Descent: Har weight update derivative (gradient) ke basis par hoti hai - 'kis direction mein weight ko move karna chahiye loss kam karne ke liye'.

Activation Functions: ReLU, Sigmoid jaise activation functions ke derivatives backpropagation mein use hote hain gradients calculate karne ke liye.

08 Real Industry Example

REAL COMPANIES

All AI companies: Training ka har single step derivatives calculate karne par based hai – yeh literally deep learning ka engine hai.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

PyTorch: `x = torch.tensor(3.0, requires_grad=True); y = x**2; y.backward(); print(x.grad)` – yeh automatically derivative calculate kar deta hai ($dy/dx = 2x = 6$).

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: Derivative sirf ek number hota hai poori function ke liye.

Sahi baat: Derivative function ke har point par alag ho sakta hai – yeh khud ek function hai (jaise $f'(x) = 2x$), single number nahi.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: Derivative kya batata hai?

A: Function kisi specific point par kitni tezi se aur kis direction mein badal raha hai.

[Intermediate] Q: Gradient descent mein derivative ka use kaise hota hai?

A: Loss function ka derivative (weight ke respect mein) yeh batata hai weight ko kis direction mein badalna chahiye taaki loss kam ho - hum derivative ke opposite direction mein move karte hain.

[Advanced] Q: Zero derivative wale points training mein kya problem create kar sakte hain?

A: Zero derivative minimum, maximum, ya saddle point ho sakta hai - agar training saddle point mein phas jaaye, gradient descent 'stuck' feel ho sakta hai kyunki update signal bahut chhota ho jaata hai.

12 Exercises

PRACTICE TIME

[Easy] $f(x) = x^2$ ka derivative kya hai?

Solution: $f'(x) = 2x$.

[Medium] $f(x) = 3x^2 + 5x$ ka derivative nikalo.

Solution: $f'(x) = 6x + 5$.

[Hard] Agar loss function ka derivative kisi weight ke respect mein zero hai, iska kya matlab ho sakta hai?

Solution: Woh weight ek local minimum, maximum, ya saddle point par ho sakta hai - training in points ke paas 'flat' region mein slow ho sakti hai.

13 Golden Summary

Concept	Easy Meaning
Derivative	Function ka rate of change kisi point par
Notation	$f'(x)$ ya df/dx
Positive/Negative	Function badh raha ya ghat raha hai
Zero Derivative	Flat point - min/max/saddle

14 One-Line Memory Trick

“Derivative batata hai pahaad kitna steep hai, is exact jagah par.”